

TEACHER KEY & RE-ENTRY GUIDE

Returning DP2 | diagnose the section profile, not merely the total

Purpose: identify whether returning students can make the complete chain: boundary -> transfer -> state change -> microscopic model, and radiative stream -> imbalance -> temperature response -> restored balance. Accept alternatives when assumptions and signs are explicit.

1 | DESCRIPTION / PSEUDO-EXPLANATION / EXPLANATION

6 MARKS

ITEM	KEY	WHY
1	D	Measured state change under a stated constraint.
2	P	Treats heat as a substance and repeats 'hotter' as 'warmer'.
3	E	Names transfer across a boundary, internal-energy change and the fixed-volume closure.
4	D	Reports two measured fluxes with directions and wavelength bands.
5	P	Uses the phenomenon's label as its own cause.
6	E	Links a rate imbalance to energy accumulation, temperature change and restored balance.

Interpretation: 0-2 indicates explanation is being judged by sentence shape; 3-4 is mixed; 5-6 is secure. Probe whether item 6 is understood as a dynamic rate balance rather than a story about stored heat.

2 | EQUATION AS GRAMMAR: SYSTEM, STATE AND PROCESS

8 MARKS

PROMPT	ACCEPTABLE ANSWER / MARKING
System and boundary [2]	System: the gas [1]. Boundary: the inner cylinder and piston faces; it is sealed, so no matter crosses [1]. Accept a broader system only if all transfers are then described consistently.
Q and W_{on} [2]	$Q = 0$ because the stated thermal insulation suppresses energy transfer by heating [1]. $W_{\text{on}} > 0$ because the piston transfers energy mechanically into the gas during compression [1].
State / process [2]	U is fixed by the thermodynamic state and has a defined change between endpoints [1]. Q and W describe pathway-dependent energy transfers during a process; they are not quantities contained in the gas [1].
Microscopic account [1]	Work done on the gas raises its internal energy; for an ideal gas the molecular energy distribution shifts upward, so mean translational kinetic energy and temperature rise.
Idealisation [1]	One assumption plus a relevant limit: negligible particle volume or intermolecular forces; weaker at high density, high pressure or near condensation.

Watch for: 'adiabatic' being translated as no energy change; Q and W being treated as stored properties; and temperature being described as energy possessed by every particle.

3 | RADIATIVE BALANCE: QUANTITIES, STREAMS AND CLOSURE

8 MARKS

PROMPT	KEY / MARKING
Albedo [1]	Dimensionless ratio of reflected short-wave radiative flux to incident short-wave radiative flux. Reject 'brightness' or an amount of radiation without a denominator.
Absorbed flux [1]	$(1 - 0.30)(340) = 238 \text{ W m}^{-2}$.
Two streams [2]	Incoming solar radiation travels towards Earth and is mainly short-wave [1]. Terrestrial emission travels outward to space and is long-wave infrared [1].
Imbalance [1]	$238 - 235 = +3 \text{ W m}^{-2}$ into the Earth system. Positive means the system gains energy per unit area per unit time.
Warming / closure [2]	Greenhouse gases absorb and re-emit selected outgoing infrared wavelengths, initially reducing the outgoing rate [1]. The positive imbalance raises system energy and temperature until outgoing long-wave emission increases to equal the absorbed solar rate [1]. Strong answers may describe a higher, colder effective emission level.
Model limit [1]	Any defensible limit: no latitude, seasons, clouds, vertical structure, spectral detail, heat capacity differences, circulation, or separate surface and atmosphere temperatures.

Precision note: keep top-of-atmosphere short-wave albedo separate from long-wave emission properties. A one-box or one-layer model is a closure scheme, not a literal atmosphere. 'Trapped' hides repeated absorption, emission and escape.

4 | LANGUAGE FORENSICS

6 MARKS

ITEM	FAMILY	MODEL REPAIR
A	substance	Energy is transferred between system and surroundings; heat is the transfer due to temperature difference, not material stored in the atmosphere.
B	agent	Entropy is a state function associated with accessible microscopic configurations; it is not an agent exerting a push.
C	attribute	Temperature is a state variable associated with thermal equilibrium and mean molecular kinetic energy for an ideal gas; it is not cargo contained by the gas.
D	process-state	At radiative equilibrium, absorbed and emitted rates are equal while radiation continues to arrive, reflect, absorb and leave.
E	model-object	The layer is a simplified representation that groups altitude, temperature and spectral structure into a few parameters.

Marking: one mark for each correct family; one mark for a repair that removes the selected confusion and states a defensible physical relationship.

5 | EXIT WARRANT

2 MARKS

[1] Equality establishes zero net radiative energy gain at that boundary and time: a steady state in the model. **[1]** Repair: *'Incoming short-wave absorption and outgoing long-wave emission continue; at steady state their rates are equal, so the modelled Earth system has no net radiative energy change.'*

PROFILE | USE THE PATTERN

30 MARKS

DOMAIN	MARKS	IF AT OR BELOW HALF
Description / explanation	/6	Require an observation sentence followed by a mechanism, assumption and evidence sentence.
System / first-law grammar	/8	Rehearse boundary -> transfer pathway -> sign -> state change before numerical problems.
Radiative balance	/8	Use two coloured streams: incoming short-wave and outgoing long-wave; quantify the imbalance before naming a temperature response.
Linguistic ontology	/6	Repair one substance, agent, attribute, equilibrium and model sentence aloud.
Epistemic warrant	/2	Separate equality of rates from cessation of processes and from causal attribution.